

The identical reasoning holds at C ($dD \parallel CS$), at D , and at every vertex: each deflection lies parallel to the radius, so the impulse always aims at S . Taking the polygon to its limit, the **radial force directions BS, CS, DS** converge continuously on the centre — the force is centripetal. Thus the equal-area observation ALONE locks the direction toward S , however the magnitude may vary with distance; combined with Proposition I this makes swept area the definitive test of a central force. *Q.E.D.*